Lecture Material #3

Information Security

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Game Theory

Network Security

- By default networks are very insecure
  -- Connected to the open internet
- There are a number of well known methods for securing a network
  - Encrypting data
  - Firewalls
  - Authentication
  - Restricted permissions
- BUT, none of the methods are perfect, and issues are common inside as well as between methods

The Problem: Network Security is Hard

Caught in an endless cycle:

Attackers hack into a system causing damage.

System Admins react to the hack by coming up with a solution to fix the exploit.

Attackers come up with a smarter way around the new solution.

Solution: Game Theory

If successful, a game theoretic approach to security can…

- Provide a mathematical framework for dealing with network security
- Can automate the job of human analyst
- Analyze hundreds of thousands of "what ifs"
- Sophisticate the decision making processes of network administrators with regard to security

Game Theory

Game Theory: A way of modeling different players choices, based on the effect of other players choices.

Player: Entity participating in the game

Action: Choice a player makes on their turn

Payoff/Reward: Gain (or loss) a player receives after choosing their action

Game Theory

Information: Games can have complete information or incomplete information. Complete means that players know the strategies and payoff of their opponents.

Bayesian Game: Game where players have incomplete information (strategies | payoffs) on the other players, but they have a probability distribution.

Nash Equilibrium: The optimal outcome of a game, where each player can receive no incremental benefit from changing actions or strategy (can be more than one).

Game Theory

We can model a "game" between an attacker, and a network administrator.

Players: Attacker, Network Administrator

Actions:

For attacker - disrupt network (ddos), plant worm, install sniffer, etc…

For network admin - add sniffer detector, remove compromised account, shut off internet traffic, etc…

Payoff:

For attacker - positive for disruption of network, stolen data. Negative for being stopped, traced….

For network admin - positive for detecting/stopping attack, normal operation. Negative for disruption, stolen data…

Identifying Attackers in a Mobile Social Network

Game Theory

Model:

- Two types of nodes, benign (user) or malicious (attacker)
- "Server" connects with nodes
- Actions for server: Nothing, Packet, surveillance
- Actions for node: Forward, Ignore, Damage
- If server does no surveillance, then malicious nodes can infiltrate network
- If server surveils everyone, the service for everyone suffers

Game Theory

Goal is to find balance

Therefore, the most compelling network security problem is to correctly define a proper operation where both types of clients are considered, and efficient defense strategies are designed with the purpose of preventing malicious activities and providing good quality services to benign nodes

Game Theory

The Game From the Server

Connect with a node, then I……

1. Do Nothing

Nobody wins - but safe I guess?

2. Send node a Packet

Normal operation - good if node is benign, bad if node is malicious

3. Set up surveillance on node

Try to catch malicious node - good if node is malicious, bad if node is benign

Game Theory

The Game From the node

Connect with server, then I……

1. Do Nothing

Nobody wins - Discard packet if received

2. Forward Packet

Normal operation - good for benign node, bad for malicious node

3. Damage Packet

Do evil things - always bad for benign node, for malicious node, good if packet, bad if surveillance

Nash Equilibrium

Who is this?

John Nash, the person portrayed in “A Beautiful Mind”

https://en.wikipedia.org/wiki/A_Beautiful_Mind_%28film%29

John Nash

- One of the early researchers in game theory
- His work resulted in a form of equilibrium named after him

Three elements in every game

- Players
  - Two or more for most games that are interesting
- Strategies available to each player
- Payoffs
  - Based on your decision(s) and the decision(s) of other(s)

Game theory: Payoff matrix

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- A payoff matrix shows the payout to each player, given the decision of each player

How do we interpret this box?

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- The first number in each box determines the payout for Person 1
- The second number determines the payout for Person 2

How do we interpret this box?

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- Example
  - If Person 1 chooses Action A and Person 2 chooses Action D, then Person 1 receives a payout of 8 and Person 2 receives a payout of 3

Back to a Core Principle: Equilibrium

- The type of equilibrium we are looking for here is called Nash equilibrium
  - Nash equilibrium: "Any combination of strategies in which each player's strategy is his or her best choice, given the other players' choices"
  - Exactly one person deviating from a NE strategy would result in the same payout or lower payout for that person

How do we find Nash equilibrium (NE)?

- Step 1: Pretend you are one of the players
- Step 2: Assume that your "opponent" picks a particular action
- Step 3: Determine your best strategy (strategies), given your opponent's action
  - Underline any best choice in the payoff matrix
- Step 4: Repeat Steps 2 & 3 for any other opponent strategies
- Step 5: Repeat Steps 1 through 4 for the other player
- Step 6: Any entry with all numbers underlined is NE

Steps 1 and 2

Person 2

Person 1	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- Assume that you are Person 1
- Given that Person 2 chooses Action C, what is Person 1's best choice?

Step 3:

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- Underline best payout, given the choice of the other player
- Choose Action B, since 12 > 10 ➔ underline 12

Step 4

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- Now assume that Person 2 chooses Action D
- Here, 10 > 8 ➔ Choose and underline 10

Step 5

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- Now, assume you are Person 2
- If Person 1 chooses A
  - 3 > 2 ➔ underline 3
- If Person 1 chooses B
  - 4 > 1 ➔ underline 4

Step 6

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- Which box(es) have underlines under both numbers?
  - Person 1 chooses B and Person 2 chooses C
  - This is the only NE

Double check our NE

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, <u>3</u>
Action B	<u>12</u>, <u>4</u>	<u>10</u>, 1

- What if Person 1 deviates from NE?
  - Could choose A and get 10
  - Person 1's payout is lower by deviating

Double check our NE

Person 2

Person 1	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- What if Person 2 deviates from NE?
  - Could choose D and get 1
  - Person 2's payout is lower by deviating

Dominant strategy

Person 2

Person 1

	Action C	Action D
Action A	10, 2	8, 3
Action B	12, 4	10, 1

- A strategy is dominant if that choice is definitely made no matter what the other person chooses
  - Example: Person 1 has a dominant strategy of choosing B

New example

Person 2

	Yes	No
Yes	20, 20	5, 10
No	10, 5	10, 10

Person 1

- Suppose in this example that two people are simultaneously going to decide on this game

New example

Person 2

	Yes	No
Yes	20, 20	5, 10
No	10, 5	10, 10

Person 1

- We will go through the same steps to determine NE

Two NE possible

Person 2

Person 1

	Yes	No
Yes	20, 20	5, 10
No	10, 5	10, 10

- (Yes, Yes) and (No, No) are both NE
- Although (Yes, Yes) is the more efficient outcome, we have no way to predict which outcome will actually occur

Two NE possible

- When there are multiple NE that are possible, economic theory tells us little about which outcome occurs with certainty

Two NE possible

- Additional information or actions may help to determine outcome
  - If people could act sequentially instead of simultaneously, we could see that 20, 20 would occur in equilibrium

Practice Problem

Alice and Bob develop two new network systems, but they forgot to agree on which security strategy to follow: Strategy 1 (ST#1) or Strategy 2 (ST#2). Their payoffs depend on which strategy they choose and whether they both apply the same one, as shown in the grid below. As always, Alice chooses a row, Bob chooses a column, and Alice's payoff is listed first:

		Bob	
		ST#1	ST#2
Alice	ST#1	10, 10	8, 4
	ST#2	4, 8	5, 5

i) Find the Nash equilibrium of this scenario, if any?

ii) Does Alice have a strictly dominant strategy? Does Bob?


Thank You!